

● EASY

● ADVANCED MATH

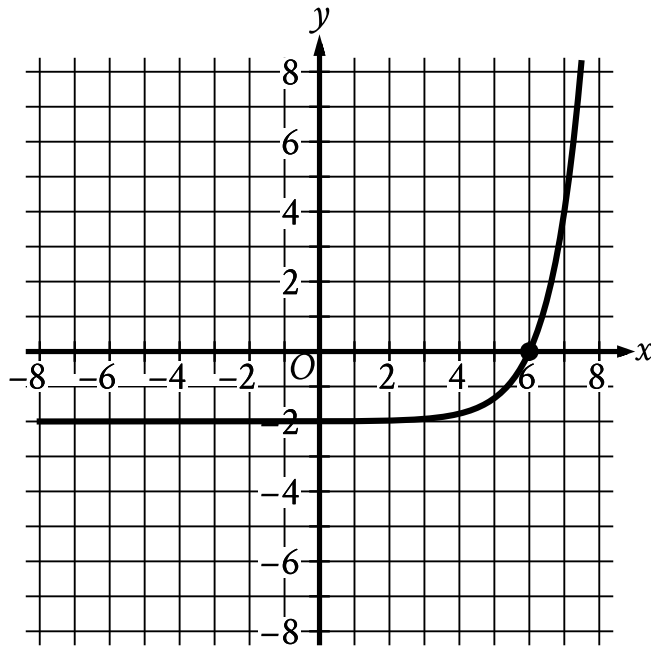
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# Nonlinear functions

10 questions · ~15 min

# 02





- Moving from left to right:
  - The curve passes from quadrant 3 to quadrant 4 to quadrant 1.
  - In quadrant 3, the curve trends up gradually to the y axis.
  - In quadrant 4, the curve trends up gradually to the point (6 comma 0).
  - In quadrant 1, the curve trends up sharply.
- As  $x$  decreases, the curve approaches the line  $y$  equals negative 2.
- The curve passes through the following points:
  - (5 comma negative four thirds)
  - (6 comma 0)

What is the  $x$ -coordinate of the  $x$ -intercept of the graph shown?

STUDENT-PRODUCED RESPONSE

|                      |                      |                      |                      |
|----------------------|----------------------|----------------------|----------------------|
| <input type="text"/> | <input type="text"/> | <input type="text"/> | <input type="text"/> |
| <input type="text"/> | <input type="text"/> | <input type="text"/> | <input type="text"/> |

|   |   |   |   |
|---|---|---|---|
| 0 | 0 | 0 | 0 |
| 1 | 1 | 1 | 1 |
| 2 | 2 | 2 | 2 |
| 3 | 3 | 3 | 3 |
| 4 | 4 | 4 | 4 |
| 5 | 5 | 5 | 5 |
| 6 | 6 | 6 | 6 |
| 7 | 7 | 7 | 7 |
| 8 | 8 | 8 | 8 |
| 9 | 9 | 9 | 9 |

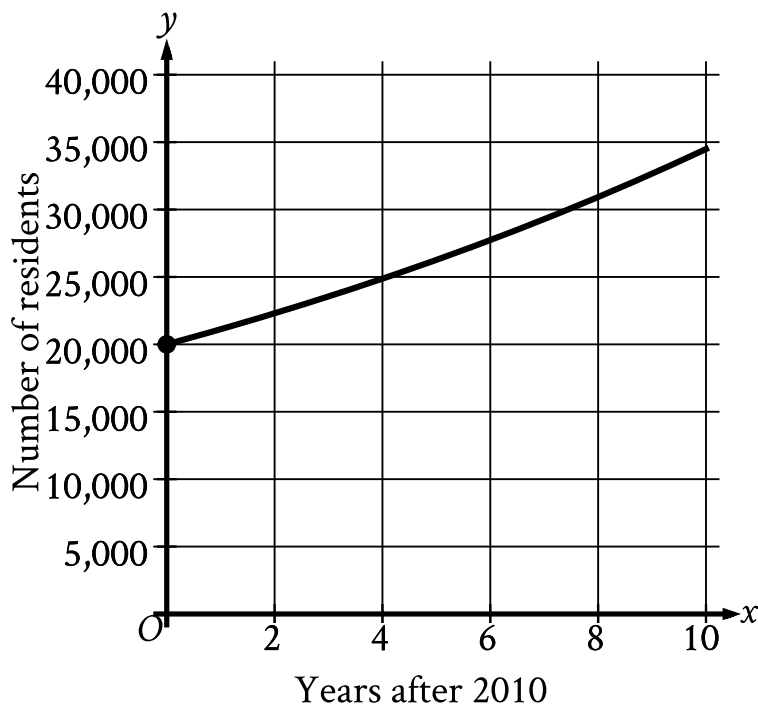
Write the answer above. Fill the matching bubble for each digit, plus the fraction bar or decimal point if needed.

**Q2**

ADVANCED MATH

The function  $f$  is defined by  $fx = 8\sqrt{x}$ . For what value of  $x$  does  $fx = 48$ ?

- A. 6
- B. 8
- C. 36
- D. 64

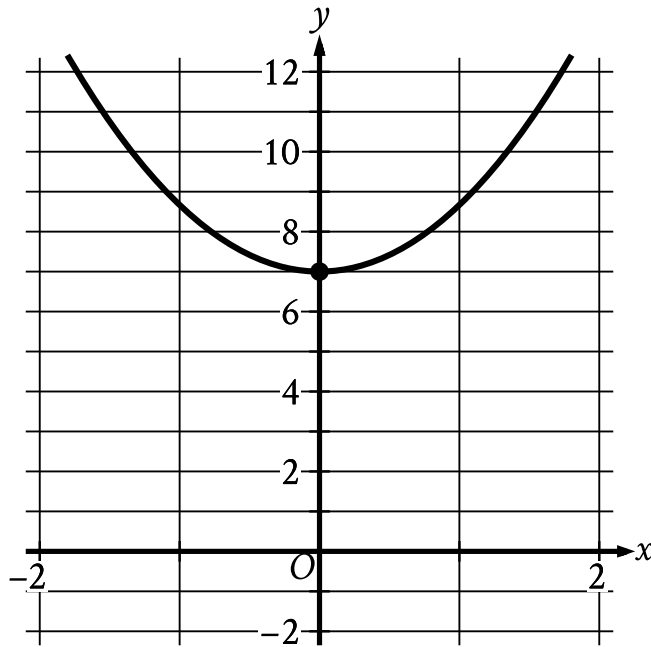


- The curve is in quadrant 1.
- The curve trends gradually up from left to right.
- The curve begins at the point (0 comma 20,000).
- The curve passes through the following points:
  - (0 comma 20,000)
  - approximately (5 comma 26,263)
  - approximately (10 comma 34,488)

The graph shown models the number of residents of a certain city  $x$  years after 2010. How many residents does this model estimate the city had in 2010?

- A. 0
- B. 2,000
- C. 20,000

D. 25,000



- The parabola opens upward.
- The vertex is at the point (0 comma 7).
- The parabola passes through the following points:
  - (negative 1 comma StartFraction 26 Over 3 EndFraction)
  - (0 comma 7)
  - (1 comma StartFraction 26 Over 3 EndFraction)

The parabola shown intersects the  $y$ -axis at the point  $x, y$ . What is the value of  $y$ ?

STUDENT-PRODUCED RESPONSE

|                      |                      |                      |                      |
|----------------------|----------------------|----------------------|----------------------|
| <input type="text"/> | <input type="text"/> | <input type="text"/> | <input type="text"/> |
| .                    | /                    | .                    | .                    |
| 0                    | 0                    | 0                    | 0                    |
| 1                    | 1                    | 1                    | 1                    |
| 2                    | 2                    | 2                    | 2                    |
| 3                    | 3                    | 3                    | 3                    |
| 4                    | 4                    | 4                    | 4                    |

|   |   |   |   |
|---|---|---|---|
| 5 | 5 | 5 | 5 |
| 6 | 6 | 6 | 6 |
| 7 | 7 | 7 | 7 |
| 8 | 8 | 8 | 8 |
| 9 | 9 | 9 | 9 |

Write the answer above. Fill the matching bubble for each digit, plus the fraction bar or decimal point if needed.

**Q5**

ADVANCED MATH

The function  $k$  is defined by  $kx = x^3 + 110$ . What is the value of  $kx$  when  $x = 2$ ?

- A. 118
- B. 116
- C. 115
- D. 110



$$P(t) = 1,800(1.02)^t$$

The function  $P$  gives the estimated number of marine mammals in a certain area, where  $t$  is the number of years since a study began. What is the best interpretation of  $P(0) = 1,800$  in this context?

- A. The estimated number of marine mammals in the area was 102 when the study began.
- B. The estimated number of marine mammals in the area was 1,800 when the study began.
- C. The estimated number of marine mammals in the area increased by 102 each year during the study.
- D. The estimated number of marine mammals in the area increased by 1,800 each year during the study.

The function  $g$  is defined by  $g(x) = \sqrt{8x + 1}$ . What is the value of  $g(3)$ ?

- A.  $\frac{5}{8}$
- B.  $\frac{25}{8}$
- C. 5
- D. 25

**Q8**

ADVANCED MATH

The function  $f$  is defined by  $fx = \frac{16}{x}$ . What is the value of  $fx$  when  $x = 17$ ?

A.  $\frac{16}{17}$

B.  $\frac{17}{16}$

C. 16

D. 17

**Q9**

ADVANCED MATH

The function  $g$  is defined by  $gx = \sqrt{x} + 300$ . What is the value of  $gx$  when  $x = 81$ ?

A. 9

B. 300

C. 309

D. 381

The function  $g$  is defined by  $gx = x^2 + 9$ . For which value of  $x$  is  $gx = 25$ ?

- A. 4
- B. 5
- C. 9
- D. 13